

CATAM: Strömgren Radius

0 Introduction

In this project, we consider the Strömgren sphere approximation and its application to different situations. We consider its accuracy for stars' H II regions compared to a model using blackbody radiation, constant temperature and density, monochromatic opacity (only at the ionisation frequency), and recombination. We then apply this model to quasars to calculate their effects on their host galaxy.

1 Question 1

$$L_\nu = 4\pi R^2 \frac{2\pi h}{c^2} \frac{\nu^3}{\exp\left(\frac{h\nu}{kT_*}\right) - 1} \quad (1)$$

To find the total luminosity, we integrate over frequency from 0 to infinity. By making the substitution $x = h\nu/k_B T_*$ so $dx/d\nu = h/kT_*$, the integral reduces to

$$L = \frac{8\pi^2 R^2 T^4 k^4}{c^2 h^3} \int_0^\infty \frac{x^3}{e^x - 1} dx \quad (2)$$

In general, this integral is solved by [1, p. 807]

$$\int_0^\infty \frac{x^{s-1}}{e^x - 1} dx = \Gamma(s)\zeta(s) \quad (3)$$

In this case $s = 4$ so integral is $\pi^4/15$, so the expression is:

$$L = \frac{2\pi^5 k^4}{15h^3 c^2} 4\pi R^2 T^4 \quad (4)$$

with $\frac{2\pi^5 k^4}{15h^3 c^2}$ being the Stefan-Boltzmann constant σ , which is given by: $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$.

$$L = \sigma 4\pi R^2 T^4 \quad (5)$$

From this, we can calculate the properties of the stars required from the two values given in the question. Giving that the three stars are *B1*, *B2*, and *G2* type main sequence stars.

$mass[M_\odot]$	Temperature [K]	Luminosity [W]	Radius [m]
1	5.800	3.9×10^{26}	6.96×10^8
7	20.000	4.0×10^{29}	1.87×10^9
12	25.000	4.0×10^{30}	3.79×10^9

Table 1: Table of properties for the 1, 7, and 12 solar mass stars

2 Question 2

Solving for integrals of this type with a non-zero lower bound (although with $s = 3$ this time) is not possible analytically, so we must find a numerical method.

$$n_{H^0} \int_{\nu_0}^{\infty} \frac{L_{\nu}}{4\pi r^2 h \nu} a_{\nu} e^{-\tau_{\nu}} d\nu = n_p n_e \alpha(T) \quad (6)$$

Substituting the form of L_{ν} for a black-body:

$$n_{H^0} \int_{\nu_0}^{\infty} \frac{1}{4\pi r^2 h \nu} 4\pi R^2 \frac{2\pi h}{c^2} \frac{\nu^3}{\exp(\frac{h\nu}{kT_*}) - 1} a_{\nu} e^{-\tau_{\nu}} d\nu = n_p n_e \alpha(T) \quad (7)$$

We use the absorption coefficient for photoionisation cross-section along with the number density of the neutral hydrogen atoms, free protons, and free electrons (assuming the medium is entirely composed of hydrogen) as follows:

$$a_{\nu} = a_{\nu_0} \left(\frac{\nu_0}{\nu}\right)^3 \quad \tau_{\nu} = \tau_{\nu_0} \left(\frac{\nu_0}{\nu}\right)^3 \quad n_p = n_e \quad n_p + n_{H^0} = n_H \quad (8)$$

Which gives:

$$n_{H^0} \int_{\nu_0}^{\infty} \frac{1}{r^2} 2\pi R^2 \frac{1}{c^2} \frac{\nu^2}{\exp(\frac{h\nu}{kT_*}) - 1} a_{\nu_0} \left(\frac{\nu_0}{\nu}\right)^3 e^{-\tau_{\nu_0} (\frac{\nu_0}{\nu})^3} d\nu = n_p^2 \alpha(T) \quad (9)$$

By making the change of variable to x and defining the ionisation fraction as:

$$x = \frac{h\nu}{kT_*}, \quad \frac{dx}{d\nu} = \frac{h}{kT_*} \quad \text{and} \quad f = \frac{n_p}{n_H}, \quad 1 - f = \frac{n_{H^0}}{n_H} \quad (10)$$

We obtain:

$$n_H (1 - f) \int_{\nu_0}^{\infty} \frac{1}{r^2} 2\pi R^2 \frac{1}{c^2} \frac{\nu^2}{e^x - 1} a_{\nu_0} \left(\frac{\nu_0}{\nu}\right)^3 e^{-\tau_{\nu_0} (\frac{\nu_0}{\nu})^3} d\nu = (n_H f)^2 \alpha(T) \quad (11)$$

Pulling the constants out of the integral, we get:

$$\frac{2\pi R^2}{r^2 c^2} \frac{1 - f}{n_H \alpha(T) f^2} a_{\nu_0} \nu_0^3 \int_{\frac{\nu_0 k T_*}{h}}^{\infty} \frac{x^2}{e^x - 1} \frac{1}{x^3} e^{-(\frac{x_0}{x})^3} dx = 1 \quad (12)$$

This integral has an infinite upper bound, which is important since we are interested in the small percentage of photons emitted above the ionisation frequency. Rather than employing a method such as Simpson's rule, which requires truncation at a finite value, we decided to use a method that doesn't need to be truncated, avoiding the error this introduces. We chose Gauss-Laguerre quadrature since the equation can easily be manipulated into the appropriate form.

To solve the integral, we manipulate it into Gauss-Laguerre form.

$$I = \int_{x_0}^{\infty} \frac{x^{-1}}{e^x - 1} e^{-\tau_{\nu_0} (x_0/x)^3} dx \quad (13)$$

Substitute $b = x - x_0$ to shift the lower bound to 0.

$$I = \int_0^{\infty} \frac{(b + x_0)^{-1}}{e^{b+x_0} - 1} e^{-\tau_{\nu_0} (x_0/(b+x_0))^3} db \quad (14)$$

$$I = \int_0^{\infty} e^{-b} \frac{(b + x_0)^{-1}}{e^{x_0} - e^{-b}} e^{-\tau_{\nu_0} (x_0/(b+x_0))^3} db \quad (15)$$

The equation is in the Gauss-Laguerre quadrature form:

$$I = \int_0^\infty e^{-x} f(x) dx \approx \sum_{i=1}^n w_i f(x_i) \quad (16)$$

where x_i is the i^{th} root of the Laguerre polynomial, and w_i is the weight given by:

$$w_i = \frac{x_i}{(n+1)^2 [L_{n+1}(x_i)]^2} \quad (17)$$

[1, p. 890] Using this, we can calculate these types of integrals faster and more accurately. We define C as:

$$C = \frac{2\pi R^2}{c^2} \frac{a_{\nu_0} \nu_0^3}{n_H \alpha(T)} \quad (18)$$

$$\frac{C}{r^2} \frac{1-f}{f^2} I = 1 \quad (19)$$

To verify that this is correct, we check dimensions:

$$[C] = \frac{[L]^2}{[LT^{-1}]^2} \frac{[L^2][T^{-1}]^3}{[L^{-3}][L^3 T^{-1}]} = [L]^2 \quad (20)$$

We see that C has units of length squared, and because the dimension of r is L , the integral has been made dimensionless.

For the initial conditions, we start at the surface of the star ($r = R_*$), with optical depth set to zero (in reality 10^{-18} as we use a log space for tau) and fully ionised ($f = 1$). In practice, we use $f = 1 - \epsilon$ to avoid a fully ionised gas in the initial step, which would divide $\Delta\tau_{\nu_0}$ by zero, setting Δr to infinity.

$$\frac{d\tau_{\nu_0}}{dr} = n_{H^0}(r) a_{\nu_0} \quad (21)$$

We can now solve this for f . To do this, we solve this quadratic to get f at each point in τ_{ν_0} space and then update r for the new $\tau_{\nu_0} + \Delta\tau_{\nu_0}$ using a standard forward Euler method. This has local error $\mathcal{O}(\Delta\tau^2)$ and global error $\mathcal{O}(\Delta\tau)$.

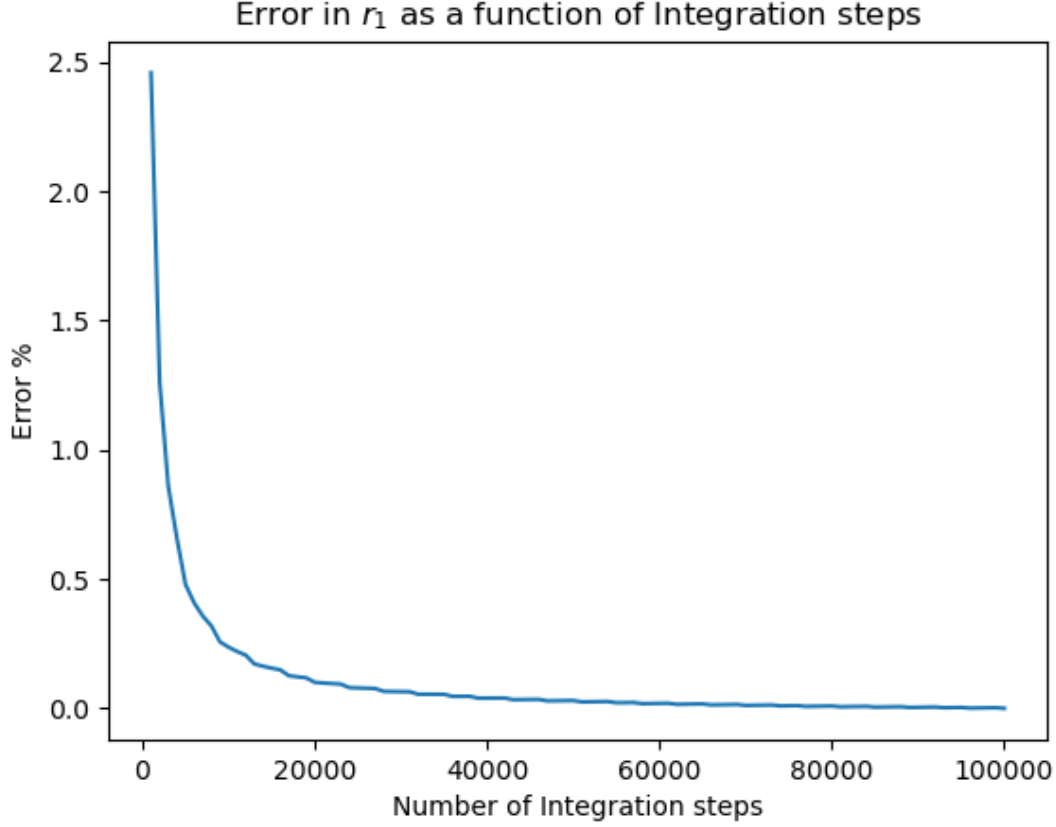


Figure 1: Error in r_1 plotted against the number of integration steps in log tau space. For the $T_* = 25,000K$, $T_{ISM} = 5000K$ case. Error is calculated against the value of r_1 with 100,000 steps.

In our case, $N = 20,000$ and so the error is 0.1%, relative to the value at $N = 100,000$. We vary the upper limit of τ for capturing the physics of the different situations: $10^{2.5}$ for the 25,000K star, $10^{2.0}$ for the 20,000K star, $10^{-1.1}$ for the 5,800K star, and $10^{5.0}$ for the quasar (although this will be somewhat less reliable due to the different L_ν function). We can assume that the error will be of roughly the same order across the different situations because, as we have varied the upper limit, the relative step size is similar. Hence, the error generalises for all our cases.

We solve in τ_{ν_0} space as it naturally results in better resolution in the region we care about, where the gas transitions from ionised to neutral. This is reinforced by Strömgren’s original paper, where he found that the transition from partially ionised to un-ionised occurs over a very small space in radius space ($1/6^{th}$ of a percent with his parameters - with significantly higher number density) [5].

The other benefit is that τ_{ν_0} is more independent of the specifics of the system (later we see r_1 in tau-space is between 3.1 and 5.3 for all the cases, ignoring the solar-type star). This becomes readily apparent from the results of the integration, which show that f decreases roughly linearly with τ_{ν_0} , thereby justifying everything we have said.

For the Gauss-Laguerre quadrature integration implementation, $N = 47$ was used throughout, as it had the lowest error from a trial of the analytic case in question 1.

$$\int_0^\infty \frac{x^3}{e^x - 1} dx = \Gamma(4)\zeta(4) = \frac{\pi^4}{15} \quad (22)$$

We found that the best N was 47 with error $1.23 \times 10^{-15} \rightarrow \pi^4/15 = 6.493939402266828$ compared to $N_{47} = 6.493939402266836$. In this specific case, $N = 27$ gives the exact value for $\pi^4/15$; however, this is due to a lucky cancellation of rounding error and for similar but not identical integrals, $N = 47$ will give

a more robust and consistent error.

Increasing N further runs into issues with floating-point arithmetic, as Python stores numbers by default to 16 significant figures. Formally, this method converges for $n \rightarrow \infty$ with error given by:

$$E = \frac{(n!)^2}{(2n)!} f^{(2n)}(\xi) \quad (23)$$

[1, p. 890] This is far more accurate than we need for our purposes.

We assume that the other integrals we will calculate, which are non-analytic and so do not lend themselves to this analysis, are of roughly the same form, so this will still hold. Even if the ideal N were to vary significantly, the error is still excellent ($error < 10^{-10}$ for $N \leq 17$) due to the rapid convergence of Gauss-Laguerre quadrature. Fluctuations in the graph are due to cumulative rounding errors, so any choice of $N \geq 20$ or so is arbitrary.

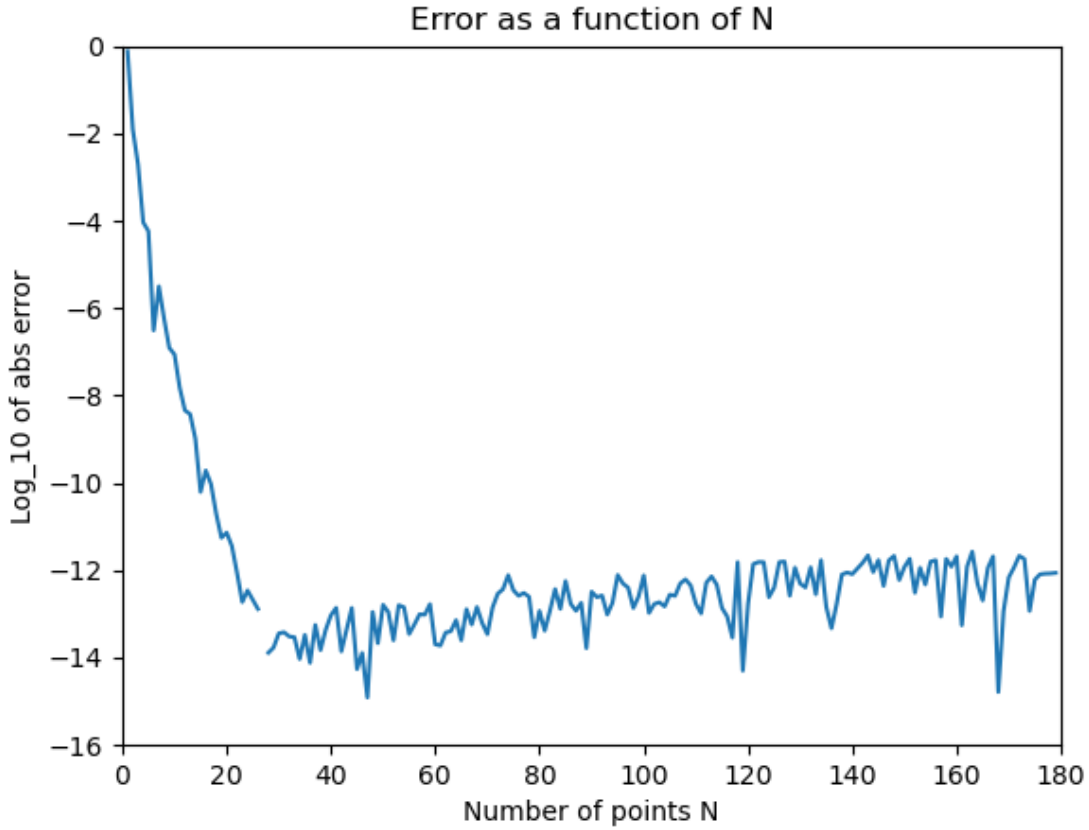


Figure 2: Gauss Laguerre quadrature error as a function of n for $\int_0^\infty \frac{x^3}{e^x - 1} dx$. The graph does not plot the "lucky" point $N=27$.

3 Question 3

3.1 Table of r_1 in parsecs for different T_* and T_{ISM}

$T_* [K]$	$T_{ISM} [K]$		
	5,000	10,000	20,000
25,000	15	18	19
20,000	5.3	6.3	6.4
5,800	4.3×10^{-4}	5.7×10^{-4}	5.8×10^{-4}

Table 2: Table of r_1 (the radius at which $f = 0.5$) in parsecs for the different combinations of Interstellar temperature and Star effective temperature

3.2 Graph of ionisation fraction against radius

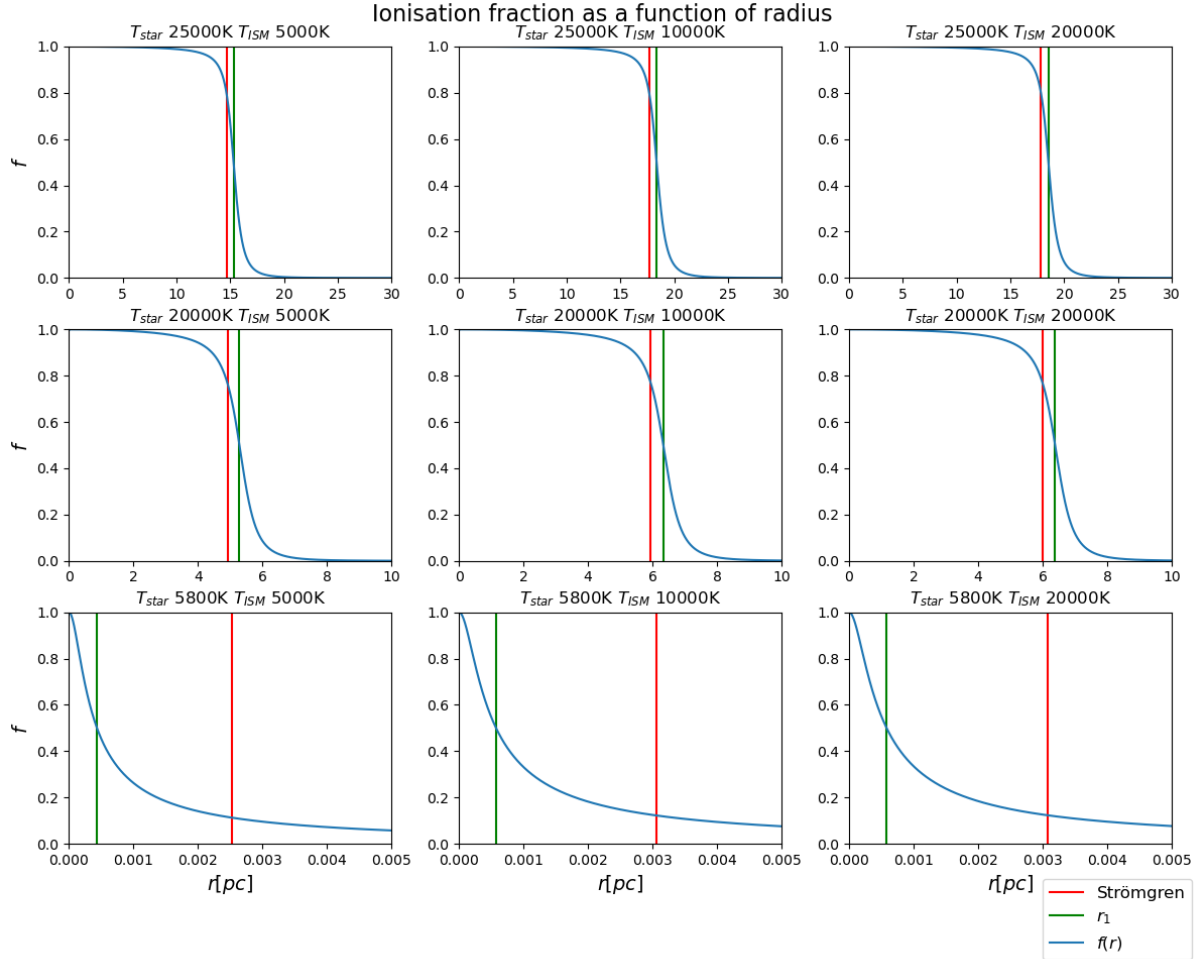


Figure 3: Ionisation fraction as a function of radius for a selection of different star and interstellar medium temperatures.

We can see from the graphs, ignoring the solar-type star for which this model breaks down, the effect of different stellar and ISM temperatures. We observe that the Strömgen radius is a slight underestimate (the numerically calculated one being roughly 4 to 7% higher). This is because the Strömgen approximation assumes a hard boundary of the sphere, whereas we see in our modelling that there is an extended

transition region. This accounts for the discrepancy.

We see that $T = 25,000$ has roughly triple the radius of $T = 20,000$ due to the strong sensitivity of Q to temperature:

$$Q(H) = \frac{4\pi}{3} r_1^3 n_H^2 \alpha_B \quad (24)$$

$$Q(H) \propto r_1^3 \quad (25)$$

Using the values of Q (tabulated later in Table 3), we see that:

$$\sqrt[3]{Q_{25,000}/Q_{20,000}} = 2.98 \quad (26)$$

Hence, this result is in agreement with the standard Strömgen equation.

A higher interstellar medium temperature decreases α_B , leading to a lower recombination rate per unit volume. This means that a higher interstellar medium temperature will result in a larger H II region. The effect of interstellar temperature is weaker, as the recombination coefficient is not particularly sensitive to temperature over the range we consider ($\alpha_B = 4.54 \rightarrow 2.52(10^{-19} m^3 s^{-1})$). From the Strömgen equation, we get:

$$\alpha_B^{-1/3} \propto r_1 \quad (27)$$

From this, we see that the difference between the highest and lowest α_B (a factor of 1.80) corresponds to the lowest α_B (highest T_{ISM}) having only a 21.6% larger r_1 as calculated by theory. From the numerical simulations, we see a difference between the lowest and highest T_* of only $7.0 \pm 0.1\%$ in r_1 for the two applicable B type stars. So simulations show an even weaker dependence on the recombination coefficient than theory predicts. Particularly, there is very little difference in α_B between the two higher temperatures (10,000K and 20,000K), which is less than 0.2% in the simulation for the two B type stars (as opposed to a predicted value of 0.9%), and this is observed in the second vs third columns.

We observe the sharp transition predicted by Strömgen [5] over a small interval in radius space of approximately one path length ($\lambda = 1/n_h a_{\nu_0}$).

By finding the spectral classifications of the stars we are asked to consider ($B1$, $B2$, $G2$), we see that the coolest star is not an O or B type star, and so the model will break down. We can see why this is true more clearly by plotting their normalised L_ν .

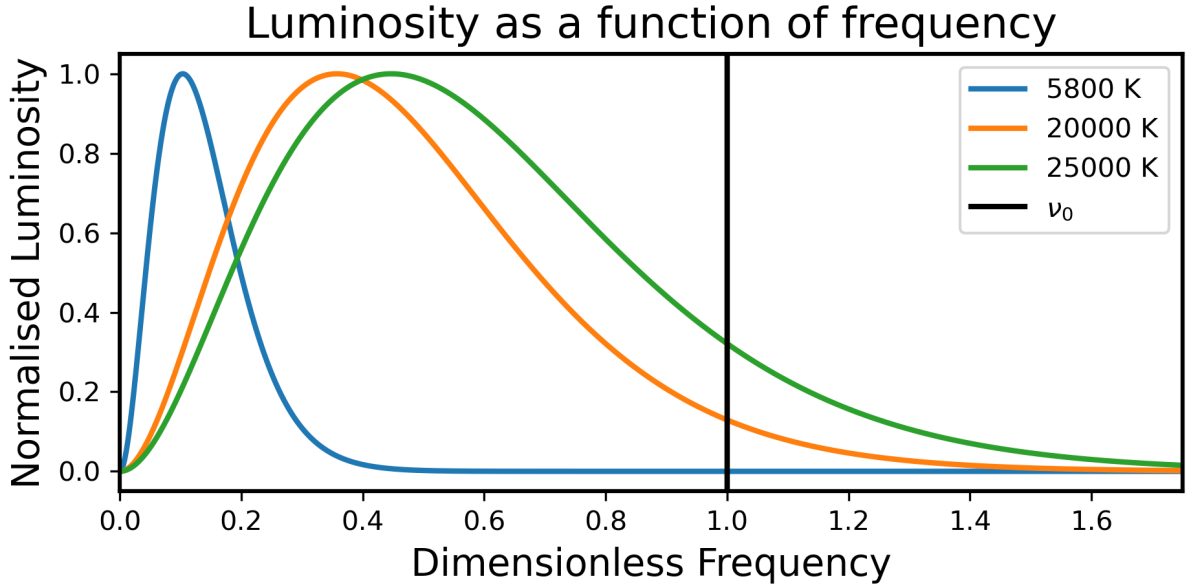


Figure 4: Normalised luminosities of the three stars we consider against frequency normalised against ν_0

From this, we can clearly see that there is a vanishingly small proportion of photons with $\nu > \nu_0$

(ionising photons) emitted from the solar-type star (in Table 3, the ionising flux for solar-type stars will be shown to be $10 \rightarrow 12$ orders of magnitude less than B type stars). Hence, the Strömgren sphere model does not apply to the solar case and justifies why it only applies for O or B type stars. This is confirmed by our simulations, which show a significant divergence between predicted and calculated values. We see an r_1 of $\sim 5 \times 10^{-4} pc = 103 AU$ vs a Strömgren radius of $5 - 6$ times that as seen in the third row in Fig.3.

3.3 Graph of Ionisation Fraction against Optical Depth

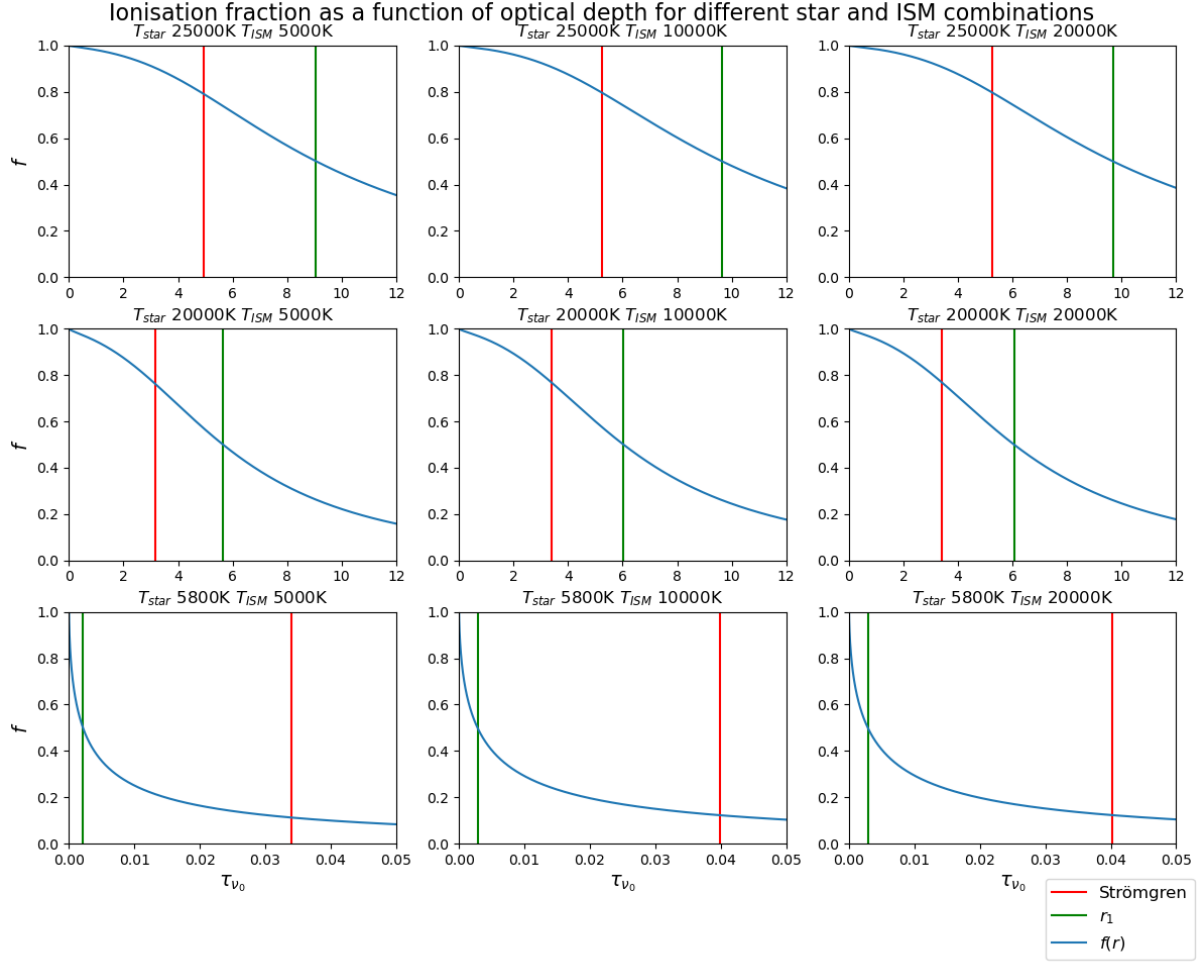


Figure 5: Ionisation fraction as a function of τ_{ν_0} for the different interstellar temperatures and star temperatures.

From these graphs, we see the benefit of using τ as the coordinate, as for the well-behaved B stars, f is roughly linear in τ . The graphs look different compared to those plotted in r -space, as the optical depths occur mostly in the partially ionised region.

4 Question 4

4.1 Strömgren approximation

$$n_{H^0} \int_{\nu_0}^{\infty} \frac{L_{\nu}}{4\pi r^2 h\nu} a_{\nu} e^{-\tau_{\nu}} d\nu = n_p n_e \alpha(T) \quad (28)$$

Integrate both sides over a spherically symmetric volume to infinity and assume that the temperature-dependent recombination coefficient $\alpha(T)$ can be approximated as: $\alpha(T) \approx \alpha_B$. Where the form of α_B is given empirically as:

$$\alpha_B(T) \approx 2.54 \times 10^{-13} Z^2 (T_4/Z^2)^{-0.8163-0.0208 \ln(T_4/Z^2)} \text{cm}^3 \text{s}^{-1} \quad (29)$$

α_B denotes case B, which is the medium being optically thick to ionising radiation such that ionising photons emitted by recombination are immediately reabsorbed, ionising another nearby neutral atom. This means that recombinations to $n = 1$ do not lead to a reduction in the net ionisation of the gas; only recombinations to $n \geq 2$ can reduce the ionisation[2, p. 139]. From now on, assuming constant temperature and metallicity, we can take α_B as constant.

$$\iiint n_{H^0} \int_{\nu_0}^{\infty} \frac{L_{\nu}}{4\pi r^2 h\nu} a_{\nu} e^{-\tau_{\nu}} d\nu d^3r = \iiint n_p n_e \alpha_B d^3r \quad (30)$$

$$\int_0^{\infty} n_{H^0} \int_{\nu_0}^{\infty} \frac{L_{\nu}}{4\pi r^2 h\nu} a_{\nu} e^{-\tau_{\nu}} d\nu 4\pi r^2 dr = \int_0^{\infty} n_p n_e \alpha_B 4\pi r^2 dr \quad (31)$$

The RHS is truncated at r_1 as we are approximating it as fully ionised within the sphere and completely un-ionised outside it. This gives the terms as $n_p n_e = n_H^2$ for $r \leq r_1$ (inside the sphere) and $n_p = n_e = 0$ for $r \geq r_1$ (outside the sphere). Evaluating the integral gives,

$$\int_0^{\infty} n_p n_e \alpha(T) 4\pi r^2 dr = n_H^2 \alpha_B \int_0^{r_1} 4\pi r^2 dr = \frac{4}{3} \pi r_1^3 n_H^2 \alpha_B \quad (32)$$

On the LHS, the $4\pi r^2$ terms cancel after changing the order of integration. We can do this due to Fubini's theorem, since the integral is positive and converges.

$$\int_{\nu_0}^{\infty} \int_0^{\infty} n_{H^0} \frac{L_{\nu}}{h\nu} a_{\nu} e^{-\tau_{\nu}} dr d\nu = \frac{4}{3} \pi r_1^3 n_H^2 \alpha_B \quad (33)$$

Substituting r for τ_{ν} we use $d\tau_{\nu}/dr = n_{H^0} a_{\nu}$ and the limits don't change as $r = 0 \rightarrow \tau_{\nu} = 0$ and $r = \infty \rightarrow \tau_{\nu} = \infty$ to get

$$\int_{\nu_0}^{\infty} \int_0^{\infty} \frac{L_{\nu}}{h\nu} e^{-\tau_{\nu}} d\tau_{\nu} d\nu = \frac{4}{3} \pi r_1^3 n_H^2 \alpha_B \quad (34)$$

Integrating over $d\tau_{\nu}$, the integral is,

$$\int_0^{\infty} e^{-\tau_{\nu}} d\tau_{\nu} = 1 \quad (35)$$

Defining $Q(H)$ as the number of ionising photons emitted per second, we get:

$$Q(H) \equiv \int_{\nu_0}^{\infty} \frac{L_{\nu}}{h\nu} d\nu = \frac{4\pi}{3} r_1^3 n_H^2 \alpha_B \quad (36)$$

as required.

4.2 Integration

$$Q = \int_{\nu_0}^{\infty} \frac{4\pi R^2 2\pi h\nu^3}{c^2 h\nu (\exp(\frac{h\nu}{kT_*}) - 1)} d\nu \quad (37)$$

$$Q = \frac{8\pi^2 R^2}{c^2} \int_{\nu_0}^{\infty} \frac{\nu^2}{\exp(\frac{h\nu}{kT_*}) - 1} d\nu \quad (38)$$

Substitute variables to get the equation into Gauss-Laguerre form.

$$x = \frac{h\nu}{kT_*} \quad \frac{dx}{d\nu} = \frac{h}{kT_*} \quad x_0 = \frac{h\nu_0}{kT_*} \quad (39)$$

$$Q = \frac{8\pi^2 R^2}{c^2} \left(\frac{kT_*}{h} \right)^3 \int_{x_0}^{\infty} \frac{x^2}{e^x - 1} dx \quad (40)$$

We substitute $b = x - x_0$ to obtain the lower bound as 0.

$$Q = \frac{8\pi^2 R^2}{c^2} \left(\frac{kT_*}{h} \right)^3 \int_0^{\infty} \frac{(x_0 + b)^2}{e^{x_0 + b} - 1} db \quad (41)$$

$$Q = \frac{8\pi^2 R^2}{c^2} \left(\frac{kT_*}{h} \right)^3 \int_0^{\infty} e^{-b} \frac{(x_0 + b)^2}{e^{x_0} - e^{-b}} db \quad (42)$$

This is in Gauss-Laguerre form.

$$\int_0^{\infty} e^{-x} f(x) dx \approx \sum_{i=1}^n w_i f(x_i) \quad (43)$$

$$f(b) = \frac{(x_0 + b)^2}{e^{x_0} - e^{-b}} \quad (44)$$

Finding Q for the three cases gives us:

$T_*[\text{K}]$	$Q[s^{-1}]$
25.000	1.76×10^{47}
20.000	6.68×10^{45}
5.800	9.06×10^{35}

Table 3: Ionising photons emitted per second for a selection of stars of different temperatures

This shows why the Strömgren sphere does not apply to solar-type stars, as they have 10 orders of magnitude lower ionising photon flux than the $B2$ star. It also shows how steeply Q varies with temperature as an increase in T of 25% leads to an increase of two orders of magnitude in ionising photon flux.

4.3 Comparing the Strömgren radius and computationally calculated r_1

$T_*[\text{K}]$	$r_{SR}[\text{pc}]$	$r_1 [\text{pc}]$
25.000	17.7	18.6
20.000	5.94	6.4
5.800	0.003 05	0.000 572

Table 4: r_1 and Strömgren radius for different stars with $T_{ISM} = 10,000\text{K}$

This is in broad agreement with how accurate we expect the Strömgren sphere to be ($4 \rightarrow 7\%$ greater is reasonable ¹), with a catastrophic error for the solar-type star, as this model doesn't apply to solar-type stars.

¹N.B. discrepancy in percentages compared to the table values arises due to rounding

5 Question 5

5.1 Quasar ionised regions

We apply the same model we have been using to the ionised region surrounding a highly luminous quasar.

$$n_{H^0} \int_{\nu_0}^{\infty} \frac{L_{\nu}}{4\pi r^2 h \nu} a_{\nu} e^{-\tau_{\nu}} d\nu = n_p n_e \alpha(T) \quad (45)$$

We use the quasar luminosity given in the question:

$$n_{H^0} \int_{\nu_0}^{\infty} 10^{24} \left(\frac{\nu}{\nu_0} \right)^{-1.4} \exp\left(-\frac{\nu}{10\nu_0}\right) \frac{1}{4\pi r^2 h \nu} a_{\nu} e^{-\tau_{\nu}} d\nu = n_p n_e \alpha(T) \quad (46)$$

Substitute the following:

$$a_{\nu} = a_{\nu_0} \left(\frac{\nu_0}{\nu} \right)^3, \quad \tau_{\nu} = \tau_{\nu_0} \left(\frac{\nu_0}{\nu} \right)^3 \quad \text{and} \quad f = \frac{n_p}{n_H}, \quad 1 - f = \frac{n_{H^0}}{n_H} \quad (47)$$

$$\frac{1-f}{n_H f^2 \alpha(T) 4\pi r^2 h} 10^{24} \int_{\nu_0}^{\infty} \left(\frac{\nu}{\nu_0} \right)^{-1.4} \exp\left(-\frac{\nu}{10\nu_0}\right) \frac{1}{\nu} a_{\nu_0} \left(\frac{\nu_0}{\nu} \right)^3 e^{-\tau_{\nu_0} \left(\frac{\nu_0}{\nu} \right)^3} d\nu = 1 \quad (48)$$

$$\frac{(1-f)a_{\nu_0}}{n_H f^2 \alpha(T) 4\pi r^2 h} 10^{24} \int_{\nu_0}^{\infty} \left(\frac{\nu}{\nu_0} \right)^{-1.4} \exp\left(-\frac{\nu}{10\nu_0}\right) \frac{1}{\nu} \left(\frac{\nu_0}{\nu} \right)^3 e^{-\tau_{\nu_0} \left(\frac{\nu_0}{\nu} \right)^3} d\nu = 1 \quad (49)$$

Manipulating the integrand into Gauss-Laguerre form, we substitute $x = \nu - \nu_0$ to get the lower bound as zero.

$$I = \int_0^{\infty} \frac{\nu_0^{4.4}}{(x + \nu_0)^{5.4}} \exp\left(-\frac{x + \nu_0}{10\nu_0} - \tau_{\nu_0} \left(\frac{\nu_0}{\nu_0 + x} \right)^3\right) dx \quad (50)$$

Substitute $b = x/10\nu_0$ in order to to extract the e^{-b} factor needed for Gauss-Laguerre form

$$I = \int_0^{\infty} \frac{\nu_0^{4.4}}{(10\nu_0 b + \nu_0)^{5.4}} \exp\left(-\frac{10\nu_0 b + \nu_0}{10\nu_0} - \tau_{\nu_0} \left(\frac{\nu_0}{\nu_0 + 10\nu_0 b} \right)^3\right) db \quad (51)$$

$$I = 10 \int_0^{\infty} \frac{1}{(10b + 1)^{5.4}} \exp\left(-b - \frac{1}{10} - \tau_{\nu_0} \left(\frac{1}{1 + 10b} \right)^3\right) db \quad (52)$$

$$I = \frac{10}{e^{1/10}} \int_0^{\infty} e^{-b} \frac{1}{(10b + 1)^{5.4}} \exp\left(-\tau_{\nu_0} \left(\frac{1}{1 + 10b} \right)^3\right) db \quad (53)$$

This is in Gauss-Laguerre form:

$$\int_0^{\infty} e^{-x} f(x) dx \approx \sum_{i=1}^n w_i f(x_i) \quad (54)$$

$$f(b) = \frac{1}{(10b + 1)^{5.4}} \exp\left(-\tau_{\nu_0} \left(\frac{1}{1 + 10b} \right)^3\right) \quad (55)$$

Taking the constants in front of I out, we get

$$\frac{a_{\nu_0}}{n_H \alpha(T) 4\pi h} 10^{24} \frac{10}{e^{1/10}} \frac{1-f}{f^2} \frac{1}{r^2} I = 1 \quad (56)$$

$$C = \frac{a_{\nu_0}}{n_H \alpha(T) 4\pi h} 10^{24} \frac{10}{e^{1/10}} \quad (57)$$

We check dimensions, noting that the 10^{24} constant given has units of WHz^{-1} as it is a luminosity per unit frequency interval

$$[C] = \frac{[L^2][ML^2T^{-2}]}{[L^{-3}][L^3T^{-1}][ML^2T^{-1}]} = [L^2] \quad (58)$$

as required. Again, we have an equation of this form, which we solve using a forward Euler method.

$$\frac{C}{r^2} \frac{1-f}{f^2} I = 1 \quad (59)$$

5.2 Quasar graphs

Then we run the same code as before with the new luminosity function and check if the radius r_1 is larger or smaller than that of the galaxy.

We assume there is no gas within $30 pc$ due to the strong quasar winds and begin integration at that radius. The offset this introduces is negligible as $r_1 \approx 100,000 pc$.

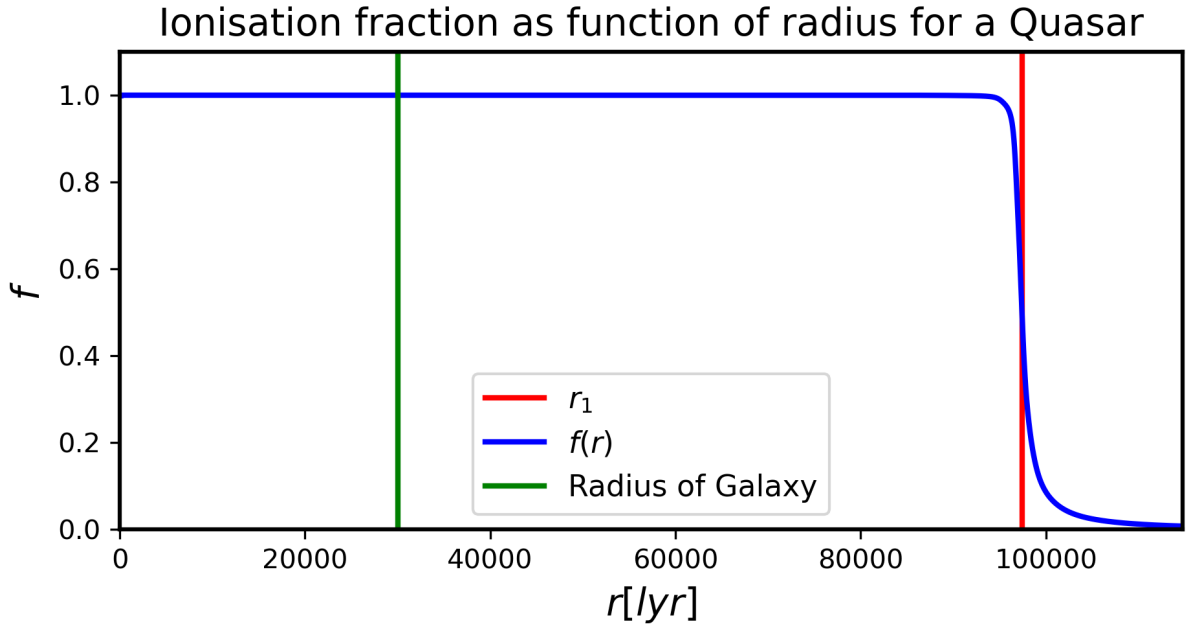


Figure 6: Ionisation fraction as a function of radius for a quasar, showing the radius with $f = 0.5$ and the stated radius of the galaxy

We see a sharper transition region in the quasar case compared to the stellar case. This is because, as the transition occurs over approximately one free path, there is a much greater number of free paths total in the quasar case (~ 2000 as opposed to ~ 10). Therefore, the transition region that occurs over one free path ($\lambda = 1/n_h a_{\nu_0}$) is fractionally smaller and so appears sharper on the graph.

There is a longer tail in f for this case due to the different L_ν function, which has a greater flux at high frequency that passes through the optically thick region, leading to partial ionisation farther beyond the sharp boundary than in the stellar case.

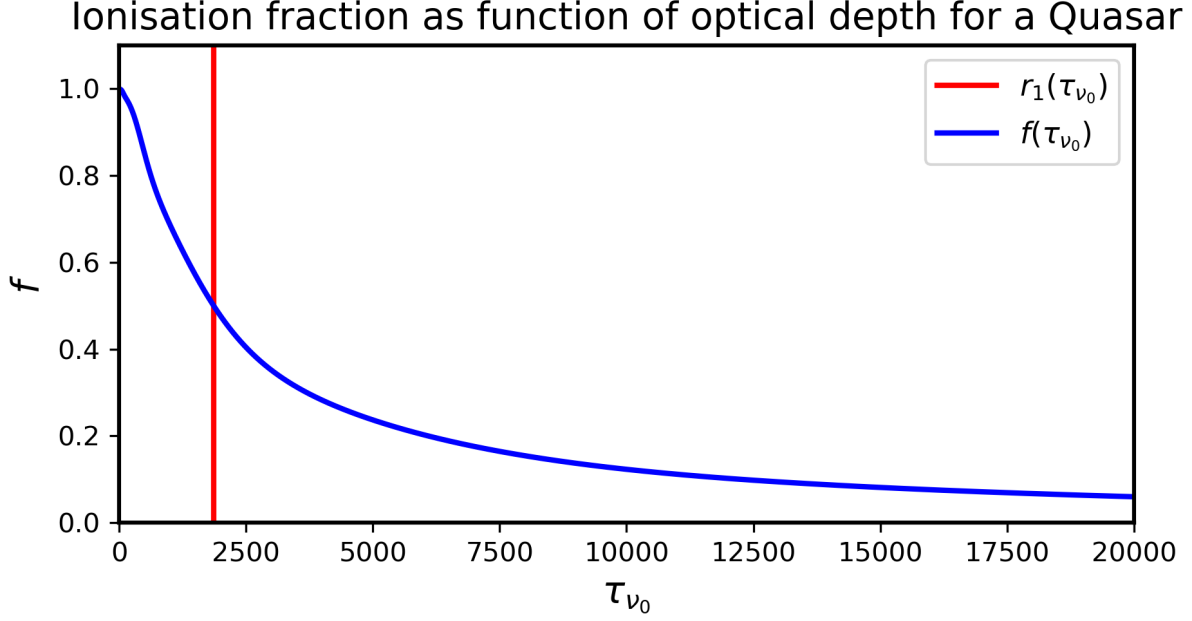


Figure 7: Ionisation fraction as a function of optical depth for a quasar.

This gives $r_1 = 97,409$ lyr for quasars, which is much greater than the stated radius of this galaxy (30,000 lyr). This means there is no neutral hydrogen within the radius of the galaxy.

For density, we are told to take a number density of $10^6 \text{ m}^{-3} = 1 \text{ cm}^{-3}$, but this is an underestimate. A true average value is closer to 2000 cm^{-3} . If we take a higher density, it would result in the galaxy not being fully ionised, as $r_1 \propto n_H^{-2/3}$ from the Strömgren equation ($Q(H) = \frac{4\pi}{3} r_1^3 n_H^2 \alpha_B$). Hence, the new r_1 would be $97,409 \times (2000)^{-2/3} = 614$ lyr, which is significantly smaller than the radius of the galaxy (30,000 lyr). So, as there is a wide range of reasonable average densities, whether the galaxy is ionised or not depends on an accurate determination of the density near the AGN. [4, p. 99]

The model will be a worse approximation for quasars than for stellar H II regions. This is because a galaxy has significantly more dust than the ISM, changing the form of the opacity terms.

6 Model issues

Strömgren estimated that the true radius, using a mean value of the absorption coefficient over the relevant frequency range of the ionising radiation, would lead to a 20 – 30% increase in the size of the ionising region, and that the effect of scattering of ionising photons emitted after recombination would be at most an increase of 15%. His method agreed with observations of H II regions in the Cygnus region.

”We find a diameter of about 200 parsecs. This agrees about as well as could be expected with the estimated diameter of the Cygnus emission regions of, say, 200 – 300 parsecs” [5]

Strömgren’s model only considers a pure monatomic hydrogen ISM. In reality, the effects of helium, metals, and dust grains increase the opacity and reduce the size of the H II region. We have assumed a constant density, metallicity, and temperature. Isothermal gas is a reasonable assumption, as the gas will be held in equilibrium by radiative cooling. We can assume metallicity will not vary significantly unless there is a high rate of mass outflow from the star. Therefore, assuming α_B is constant is reasonable. The model assumes spherical symmetry and uniform density. In reality, ISM gas is inhomogeneous or ‘clumpy’, which allows ‘leaking’ of photons. This leads to the irregular shapes of H II regions we observe, which reduces the size of the ionised region. Hence, uniform density is not an accurate assumption, but it is sufficient to illustrate the general behaviour.

We have assumed a monochromatic opacity, only including the effects of photoionisation. For a pure monatomic hydrogen ISM, this is a good model; however, real ISMs will have dust, helium, and metals, all of which will absorb ionising photons and reduce the size of the ionised region.

Real stars are not perfect blackbodies; in the B type stars we consider, there is significant absorption of ionising photons by atmospheric hydrogen. In more accurate stellar models [3] a $B1$ star has a $Q \approx 10^{45} \text{ s}^{-1}$ showing our value of $1.76 \times 10^{47} \text{ s}^{-1}$ to be a significant overestimate. The numerical approximations we used have intrinsic errors. We showed that the Gauss-Laguerre method is highly accurate ($\text{error} < 10^{-15}$) and that the global error of the forward Euler method is $\approx 0.1\%$, so negligible compared to the other approximations we made.

7 Bibliography

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